AhmedNLP at IslamicEval 2026: Learning Answer Relevance from Within-Response Citation Preferences

Ahmed Almatrafi
College of Computing and Information Systems
Islamic University of Madinah, Saudi Arabia
almatrfi.ahmed@gmail.com

Abstract

IslamicEval 2026 Subtask 4 asks whether a Qur'anic or Hadith citation is useful for answering an Arabic question, rather than merely related to its topic. We treat this decision as ranking under shared response context. An Arabic instruction prompt is scored by the first answer-token logit difference between *na'am* ("yes") and *la* ("no"). We adapt ALLaM 7B with LoRA and a response-balanced RankNet loss over 1,063 positive–negative pairs drawn from 348 mixed-label responses. Inference uses logits and one calibrated threshold without generation. On Development, the frozen checkpoint obtained 0.87 Question Macro-F1 under question-grouped out-of-fold (OOF) threshold calibration, compared with 0.78 for zero-shot ALLaM. AhmedNLP_AllamR achieved 0.77 on the official test, our best of three runs. The largest Development-to-test drop occurred on all-negative questions, motivating direct supervision for groups containing only non-relevant citations.

1 Introduction

A citation can be authentic yet unhelpful. A Qur'anic verse or Hadith may share the subject of a question without stating a ruling, supplying a necessary premise, or otherwise supporting the answer. IslamicEval 2026 Subtask 4, *Answer Relevance*, makes this distinction explicit: topical similarity alone is insufficient (Alharbi et al., 2026). This extends the IslamicEval line of work on grounded Islamic question answering and hallucination analysis (Mubarak et al., 2025). It also differs from passage-retrieval tasks such as Qur'an QA, where the main problem is finding answer-bearing passages from a collection (Malhas et al., 2023).

Our first systems classified each citation independently. They ranked many examples sensibly, but their final decisions remained sensitive to class imbalance and the score threshold. We then noticed that training rows retain a `Response_ID`: citations extracted from the same generated response share a question and generation context. When one such citation is labeled relevant and another non-relevant, their relation supplies a direct preference: the former should receive a higher score. This observation changed the learning unit from an isolated citation to a within-response pair.

The resulting system combines an Arabic backbone, parameter-efficient adaptation, and a pairwise objective. Its main contribution is not a new loss, but a task-specific alignment between data structure and supervision. The experiments support three observations: (1) response-balanced pair construction provided cleaner supervision than pointwise classification; (2) first-token Arabic label logits allowed deterministic scoring without free generation; and (3) question-grouped threshold calibration exposed both the gain of pairwise training and its blind-test weakness on all-negative questions.

2 Related Work

Our scoring interface follows generative rerankers that interpret label-token logits as relevance scores. MonoT5 generates relevance labels and shows that the target-token choice itself affects ranking quality (Nogueira et al., 2020). FIRST replaces autoregressive list generation with first-token logits and a learning-to-rank loss (Gangi Reddy et al., 2024). We transfer the scoring principle, but use Arabic binary labels and response-local pairs rather than listwise document identifiers.

The supervision objective builds on RankNet's pairwise logistic loss (Burges et al., 2005). We also tested LambdaLoss, which connects pair weighting to ranking metrics (Wang et al., 2018), but it did not outperform response-balanced RankNet under the matched metric. MAIR reports that instruction-tuned retrievers and rerankers remain weak on some long-tail tasks (Sun et al., 2024),

Proceedings of The Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026) - Volume 2 Shared Tasks, pages 1045–1049
October 28-29, 2026 ©2026 Association for Computational Linguistics

	Train	Development
Citations	6,231	282
Questions	496	111
Responses	1,652	122
Relevant	5,254	151
Non-relevant	977	131

Table 1: Task 4 data used in our experiments.

while FollowIR finds that existing retrieval models often reduce detailed instructions to basic keywords (Weller et al., 2025). BRIGHT shows a related failure of standard retrieval when relevance requires reasoning beyond surface semantic matching (Su et al., 2025). These results motivated our explicit evidence-usefulness instruction and our direct evaluation of multilingual cross-encoders.

For Arabic, we use ALLaM (Bari et al., 2025). HalluTruthQA, a contemporaneous Arabic preprint, evaluates ALLaM and Falcon-H1 among four models and reports that no single model leads across detection, localization, factual verification, and explanation (Bouchekif et al., 2026). We therefore evaluated Falcon-H1 (Zuo et al., 2025) as a separate backbone. Neither Falcon nor the off-the-shelf reranker branch displaced ALLaM on the official metric.

3 Task and Data

Each row contains an Arabic question, a citation span, its type (Qur'anic *Ayah* or Hadith *matn*), and a generated-response identifier. The output is 1 if the citation answers the question or supplies necessary evidence; otherwise it is 0. The official metric averages positive-class F1 across questions. For an all-negative question, F1 is 1 only if every prediction is negative, and 0 otherwise. We call this *Question Macro-F1*.

Table 1 summarizes the labeled splits. Train is positive-skewed. Within each `Response_ID`, we pair every relevant citation with every non-relevant citation: the Cartesian product (e.g., $2 \times 3 = 6$ pairs). The 348 *mixed-label responses*, containing both labels, yield 1,063 pairs from 1,267 of 6,231 citations. The remaining 1,304 *homogeneous responses* contain only one label. They cannot form pairs but remain in evaluation. With 1–16 pairs per response, unweighted pairs would overrepresent larger groups.

4 System

4.1 Prompt and deterministic score

We use the 7B instruction version of ALLaM, an Arabic–English model from the ALLaM family (Bari et al., 2025).[1] The prompt defines relevance, includes only the question and candidate citation, and ends with the model's native chat template. Early checks showed that asking for Arabic *na'am/la* was more stable than asking for the abstract symbols 1/0. The latter changed both tokenization and the learned conversational interface; the two label formats were therefore not interchangeable. This is consistent with prior evidence that relevance-label tokens are not neutral design choices (Nogueira et al., 2020).

For question q and citation c, z_{yes} and z_{no} are the raw, pre-softmax logits for Arabic *na'am* and *la* at the first answer position:

$$s(q, c) = z_{\text{yes}}(q, c) - z_{\text{no}}(q, c). \tag{1}$$

Following first-token reranking (Gangi Reddy et al., 2024), we use a forward pass without calling `generate()`. Generation defaults such as temperature, top-p, and top-k therefore do not affect these logits; no token is sampled.

4.2 Response-balanced RankNet

Let c^+ and c^- be relevant and non-relevant citations from the same response. We optimize the RankNet logistic preference loss (Burges et al., 2005):

$$\mathcal{L}_{\text{pair}} = \log\big(1 + \exp\big[-(s(q, c^+) - s(q, c^-))\big]\big)\,. \tag{2}$$

If response r produces n_r pairs, each pair receives weight proportional to $1/n_r$. Thus, a response with 16 pairs does not contribute 16 times as much as a response with one pair. This weighting keeps the shared response, rather than the number of citation pairs, as the effective supervision unit.

4.3 Adaptation and decoder

LoRA (Hu et al., 2022) is applied to the query, key, value, and output projections with rank 8, $\alpha = 16$, and dropout 0.05. This exposes 8,388,608 trainable parameters, or 0.12% of the model. We train for 50 optimizer steps with learning rate 2×10^{-5}, pair batch size 2, gradient accumulation 4, and bfloat16 arithmetic. Maximum input length is 1,024 tokens.

[1] Official checkpoint model card.



Figure 1: The same deterministic scorer supports pairwise training and threshold-only inference. Test labels are produced automatically by the frozen global threshold, without generation, sampling, or manual intervention.

System	OOF	Fixed Dev
Qwen 4B + LambdaLoss	0.78	0.78
ALLaM zero-shot	0.78	0.81
ALLaM + pointwise BCE	0.77	0.81
ALLaM + response RankNet	**0.87**	**0.88**

Table 2: Question Macro-F1 on Development. OOF refers to question-grouped threshold calibration.

Post-core branch	OOF F1	Δ
Response RankNet	**0.87**	–
16-feature stacked calibrator	0.86	-0.01
Ayah/matn thresholds	0.85	-0.01
Null-anchor relative score	0.84	-0.03

Table 3: Later branches versus frozen response RankNet. Differences are computed before rounding.

At inference, the frozen adapter scores rows in batches of four. A citation is relevant if $s(q, c) \geq \tau$. We retained one global threshold because response-relative margins, largest-gap rules, type-specific thresholds, and learned calibrators either reduced held-out F1 or produced negligible full-Development gains. The primary submitted run used $\tau = -4.0$.

5 Experiments and Results

5.1 Validation protocol

The adapter was trained once on Train. *Out-of-fold (OOF)* evaluates each question using a threshold chosen without it: we group Development by `question_id` into five folds, calibrate on four, and evaluate the fifth. This is *threshold-calibration OOF*, without model retraining. Fixed Dev applies the median fold threshold to all Development rows. OOF controls threshold selection, but not repeated model and prompt choices on these 111 questions.

5.2 Development comparison

Table 2 reports decision-relevant branches under the same official metric. Increasing capacity was not sufficient: Qwen3 rerankers, available at several scales including 4B and 8B (Zhang et al., 2025), and a Falcon-H1 branch remained below the selected system. Pointwise BCE also failed to improve over ALLaM zero-shot. Response-balanced pairwise adaptation raised frozen-checkpoint OOF F1 from 0.78 to 0.87. The best score observed during training on the full Development set was 0.88; we keep it separate from the more conservative frozen-checkpoint estimate.

We organized the remaining experiments by the hypothesis they tested. LambdaLoss tested metric-aware pair weighting. Qwen 8B and Falcon-H1 tested capacity and backbone alignment. Five multilingual cross-encoders, including mGTE (Zhang et al., 2024), tested transfer from general retrieval. The null anchor tested an explicit "no suitable citation" boundary, while stacked, type-aware, and fusion branches tested post-hoc calibration. None exceeded the selected RankNet checkpoint under a matched held-out protocol. The strongest generic cross-encoder reached 0.60 OOF F1, suggesting that ordinary search relevance did not transfer directly to evidence usefulness. Several branches improved ROC-AUC or full-Development F1 while reducing OOF Question Macro-F1; we rejected them because the official metric determined system selection.

5.3 Decision gates and reliability checks

We used the response RankNet OOF score as a gate for later components. Table 3 lists representative alternatives evaluated after the core checkpoint. The stacked calibrator used 16 score, rank, group-size, and citation-type features with nested tuning. It improved one outer fold but reduced the aggregate score. Separate Ayah and matn thresholds produced a full-Development gain below 10^{-3} while losing 0.01 OOF F1. The null-anchor model created a synthetic "no suitable citation" candidate to incorporate homogeneous groups; its relative score was better than its raw score, but still below the response-pair baseline. These failures support a narrow conclusion: Development was too small to justify extra learned boundaries.

Four folds selected -4.0 and one selected -3.75, with held-out F1 of 0.92, 0.85, 0.92, 0.90, and 0.75 for folds 0–4. The full-Development optimum, -4.0625, gives the same labels as -4.0. We fixed inference batch size at four because bfloat16

Run	τ	Pred. 1/0	Test F1
AllamR	-4.000	510/255	**0.77**
AllamM	-3.875	506/259	0.76
AllamC	-3.750	499/266	0.76

Table 4: Official Question Macro-F1 for the last three valid submissions.

Question type	Dev	Test	Δ
All Negative	0.85	0.60	-0.25
All Positive	0.96	0.86	-0.10
Mixed	0.83	0.72	-0.11

Table 5: Post-hoc F1 by question label composition. Dev uses the best training-time evaluation, not OOF; Test uses the frozen submitted system.

scores changed slightly after adapter reload and with batch composition. The prompt, model, length cap, batch size, score, and threshold jointly specify deployment.

5.4 Official runs and error analysis

The organizers evaluated our final three valid runs on 765 citations from 271 questions. The runs used the same frozen model and differed only in their decision boundary (Table 4). AhmedNLP_AllamR achieved 0.77 and was best among our three runs. We do not infer an overall competition rank from these results.

AllamR lost 0.10 F1 from OOF to test. All-negative questions declined most: 0.25 F1, over twice the other groups (Table 5). On the official test, AllamR produced 442 true positives, 68 false positives, 82 false negatives, and 173 true negatives. The corresponding row-level positive-class F1 was 0.85 and accuracy was 0.80. These values are diagnostic only; the submitted score remains Question Macro-F1. On Development, the selected training-time evaluation produced 23 false positives and 14 false negatives. Error counts were similar for Ayah (12/6) and matn (11/8), so citation type alone did not explain the failures. Table 6 grounds the two recurring patterns in observed errors.

The all-negative decline is consistent with RankNet's supervision gap: it orders mixed-response citations but never directly learns to reject an entire negative group. One false positive changes an all-negative question's F1 from 1 to 0. Repeated selection on 111 Development questions may also contribute; threshold-only OOF does not control it. These are plausible explanations, not proven causes. The three thresholds differ by less than 0.01 test F1, so small boundary changes do not resolve the gap. The null anchor covered homogeneous groups but reduced held-out F1; balanced direct negative-group supervision remains future work.

Question	Citation	G/P
ما حكم بيع الدم	وَلَا تَقْتُلُوا النَّفْسَ الَّتِي حَرَّمَ اللهُ إِلَّا بِالْحَقِّ	0/1
ما هو تعريف الحرام	نَهَى رَسُولُ اللهِ صَلَّى اللهُ عَلَيْهِ وَسَلَّمَ عَنْ كُلِّ ذِي نَابٍ مِنَ السِّبَاعِ ، وَعَنْ كُلِّ ذِي مِخْلَبٍ مِنَ الطَّيْرِ .	1/0

Table 6: Representative Development errors in the original Arabic. The first is a topical false positive; the second is a false negative requiring an additional inference. G/P denotes gold/prediction.

6 Conclusion

AhmedNLP used an Arabic first-token scorer, LoRA, response-balanced RankNet, and one global threshold for IslamicEval 2026 Answer Relevance. It achieved 0.87 frozen-checkpoint OOF F1 and 0.77 official test F1, our best submission. Within-response pairs outperformed larger backbones and extra decoder stages. Future work should combine this relative signal with direct supervision from homogeneous negative responses.

Code availability. Supplementary material includes training code, the Arabic prompt, the selected adapter, scoring, and calibration artifacts.

Limitations

Development has only 111 questions; repeated selection may overstate generalization. OOF holds out questions only for threshold calibration. We used one main seed and do not claim seed robustness. Pairwise supervision excludes homogeneous responses. Post-submission error analysis cannot establish the cause of the generalization gap.

References

Rahaf Alharbi, Abdulelah Alturki, Watheq Mansour, Rana Malhas, Hamdy Mubarak, Kareem Darwish, Tamer Elsayed, and Walid Magdy. 2026. IslamicEval 2026: The second shared task of capturing LLMs hallucination in Islamic content. In *Proceedings of the Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026)*.

M. Saiful Bari, Yazeed Alnumay, Norah A. Alzahrani, Nouf M. Alotaibi, Hisham A. Alyahya, Sultan

AlRashed, Faisal A. Mirza, Shaykhah Z. Alsubaie, Hassan A. Alahmed, Ghadah Alabduljabbar, Raghad Alkhathran, Yousef Almushayqih, Raneem Alnajim, Salman Alsubaihi, Maryam Al Mansour, Saad Amin Hassan, Majed Alrubaian, Ali Alammari, Zaki Alawami, and 7 others. 2025. ALLaM: Large language models for arabic and english. In *International Conference on Learning Representations*.

Abdessalam Bouchekif, Mohammed-En-Nadhir Zighem, Salah Eddine Bekhouche, Hichem Telli, Somaya Eltanbouly, Shahd Gaben, Heba Sbahi, Samer Rashwani, Mutaz Al-Khatib, Emad Mohamed, Mohammed Ghaly, and Abdenour Hadid. 2026. HalluTruthQA: A fine-grained benchmark for hallucination detection, localization, and explanation in Arabic question answering. *arXiv preprint arXiv:2607.20219*.

Christopher J. C. Burges, Tal Shaked, Erin Renshaw, Ari Lazier, Matt Deeds, Nicole Hamilton, and Gregory N. Hullender. 2005. Learning to rank using gradient descent. In *Proceedings of the 22nd International Conference on Machine Learning*, pages 89–96.

Revanth Gangi Reddy, JaeHyeok Doo, Yifei Xu, Md Arafat Sultan, Deevya Swain, Avirup Sil, and Heng Ji. 2024. FIRST: Faster improved listwise reranking with single token decoding. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 8642–8652, Miami, Florida, USA. Association for Computational Linguistics.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*.

Rana Malhas, Watheq Mansour, and Tamer Elsayed. 2023. Qur'an QA 2023 shared task: Overview of passage retrieval and reading comprehension tasks over the holy Qur'an. In *Proceedings of ArabicNLP 2023*, pages 690–701, Singapore (Hybrid). Association for Computational Linguistics.

Hamdy Mubarak, Rana Malhas, Watheq Mansour, Abubakr Mohamed, Mahmoud Fawzi, Majd Hawasly, Tamer Elsayed, Kareem Mohamed Darwish, and Walid Magdy. 2025. IslamicEval 2025: The first shared task of capturing LLMs hallucination in islamic content. In *Proceedings of The Third Arabic Natural Language Processing Conference: Shared Tasks*, pages 480–493, Suzhou, China. Association for Computational Linguistics.

Rodrigo Nogueira, Zhiying Jiang, Ronak Pradeep, and Jimmy Lin. 2020. Document ranking with a pre-trained sequence-to-sequence model. In *Findings of the Association for Computational Linguistics: EMNLP 2020*, pages 708–718, Online. Association for Computational Linguistics.

Hongjin Su, Howard Yen, Mengzhou Xia, Weijia Shi, Niklas Muennighoff, Han-yu Wang, Haisu Liu, Quan Shi, Zachary S. Siegel, Michael Tang, Ruoxi Sun, Jinsung Yoon, Sercan O. Arik, Danqi Chen, and Tao Yu. 2025. BRIGHT: A realistic and challenging benchmark for reasoning-intensive retrieval. In *International Conference on Learning Representations*.

Weiwei Sun, Zhengliang Shi, Wu Jiu Long, Lingyong Yan, Xinyu Ma, Yiding Liu, Min Cao, Dawei Yin, and Zhaochun Ren. 2024. MAIR: A massive benchmark for evaluating instructed retrieval. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 14044–14067, Miami, Florida, USA. Association for Computational Linguistics.

Xuanhui Wang, Cheng Li, Nadav Golbandi, Michael Bendersky, and Marc Najork. 2018. The LambdaLoss framework for ranking metric optimization. In *Proceedings of the 27th ACM International Conference on Information and Knowledge Management*, pages 1313–1322.

Orion Weller, Benjamin Chang, Sean MacAvaney, Kyle Lo, Arman Cohan, Benjamin Van Durme, Dawn Lawrie, and Luca Soldaini. 2025. FollowIR: Evaluating and teaching information retrieval models to follow instructions. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 11926–11942, Albuquerque, New Mexico. Association for Computational Linguistics.

Xin Zhang, Yanzhao Zhang, Dingkun Long, Wen Xie, Ziqi Dai, Jialong Tang, Huan Lin, Baosong Yang, Pengjun Xie, Fei Huang, Meishan Zhang, Wenjie Li, and Min Zhang. 2024. mGTE: Generalized long-context text representation and reranking models for multilingual text retrieval. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing: Industry Track*, pages 1393–1412, Miami, Florida, US. Association for Computational Linguistics.

Yanzhao Zhang, Mingxin Li, Dingkun Long, Xin Zhang, Huan Lin, Baosong Yang, Pengjun Xie, An Yang, Dayiheng Liu, Junyang Lin, Fei Huang, and Jingren Zhou. 2025. Qwen3 Embedding: Advancing text embedding and reranking through foundation models. *arXiv preprint arXiv:2506.05176*.

Jingwei Zuo, Maksim Velikanov, Ilyas Chahed, Younes Belkada, Dhia Eddine Rhayem, Guillaume Kunsch, Hakim Hacid, Hamza Yous, Brahim Farhat, Ibrahim Khadraoui, Mugariya Farooq, Giulia Campesan, Ruxandra Cojocaru, Yasser Djilali, Shi Hu, Iheb Chaabane, Puneesh Khanna, Mohamed El Amine Seddik, Ngoc Dung Huynh, and 8 others. 2025. Falcon-H1: A family of hybrid-head language models redefining efficiency and performance. *arXiv preprint arXiv:2507.22448*.